Week 6 Tutorial Solutions

Question 24

Oil is leaking out of a yacht damaged at sea. The leakage rate increases each hour, as recorded in the following table:

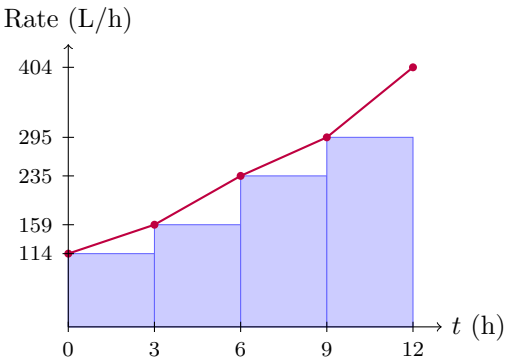
Time (h)	0	3	6	9	12
Oil Leakage Rate (L/h)	114	159	235	295	404

Using a pair of Riemann sums, find reasonable upper and lower estimates of the total quantity of oil that has leaked out over the 12 hours.

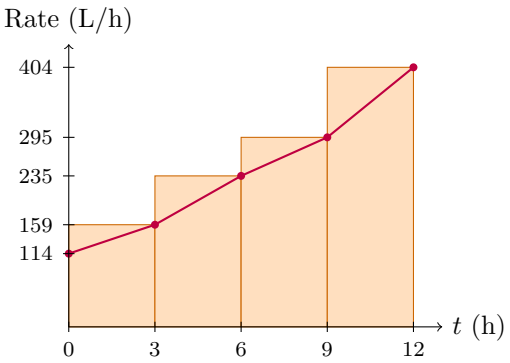
Solution

The leakage rate is increasing, so a left sum underestimates the total and a right sum overestimates it. With $n = 4$ subintervals on $[0, 12]$,

$$\Delta t = \frac{12 - 0}{4} = 3 \text{ h}.$$



Left sum (lower estimate)



Right sum (upper estimate)

Lower estimate (left sum):

$$\begin{aligned} L_4 &= 3(114) + 3(159) + 3(235) + 3(295) \\ &= 3(114 + 159 + 235 + 295) \\ &= 3(803) \\ &= 2409 \text{ L} \end{aligned}$$

Upper estimate (right sum):

$$\begin{aligned} R_4 &= 3(159) + 3(235) + 3(295) + 3(404) \\ &= 3(159 + 235 + 295 + 404) \\ &= 3(1093) \\ &= 3279 \text{ L} \end{aligned}$$

Conclusion: The amount of oil leaked over the 12 hours lies between 2409 L and 3279 L.

Question 25

Find the exact value of the integral

$$\int_{-3}^{0} \left(1 + \sqrt{9 - x^2}\right) dx$$

by interpreting it in terms of areas.

Solution

Split the integral into two parts:

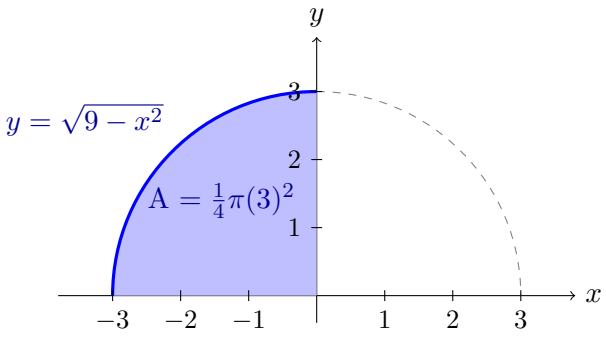
$$\int_{-3}^{0} \left(1 + \sqrt{9 - x^2}\right) dx = \int_{-3}^{0} 1\, dx + \int_{-3}^{0} \sqrt{9 - x^2}\, dx.$$

First integral. The graph of $y = 1$ is a horizontal line, so the area beneath it is a rectangle of height 1 and width 3 (from $x = -3$ to $x = 0$):

$$\int_{-3}^{0} 1\, dx = (3)(1) = 3.$$

Second integral. Rearranging $y = \sqrt{9 - x^2}$ gives $x^2 + y^2 = 9$, which is a circle of radius 3. Since $y \geq 0$, this is the upper semicircle, and over $-3 \leq x \leq 0$ the area under the curve is a quarter circle of radius 3:

$$\int_{-3}^{0} \sqrt{9 - x^2}\, dx = \frac{1}{4}\pi r^2 = \frac{1}{4}\pi(3)^2 = \frac{9\pi}{4}.$$



Combining:

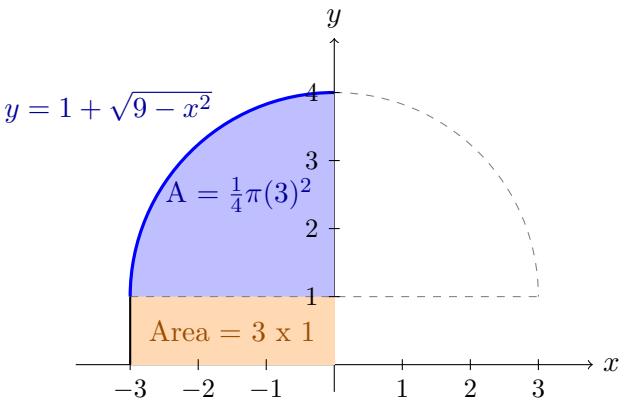
$$\int_{-3}^{0} \left(1 + \sqrt{9 - x^2}\right) dx = 3 + \frac{9\pi}{4} \approx 10.07.$$

Alternative Solution

Instead of splitting the integrand, interpret the whole function $y = 1 + \sqrt{9 - x^2}$ geometrically. Since $y - 1 = \sqrt{9 - x^2}$ gives $x^2 + (y - 1)^2 = 9$, the graph is the upper semicircle of radius 3 centered at $(0, 1)$, i.e. the semicircle $y = \sqrt{9 - x^2}$ shifted up by 1 unit.

On $-3 \leq x \leq 0$, the area under this curve (down to the x-axis) consists of two pieces:

- a **quarter circle** of radius 3 (centered at $(0, 1)$), with area $\frac{1}{4}\pi(3)^2 = \frac{9\pi}{4}$;
- a **rectangle** below it, between $y = 0$ and $y = 1$, of width 3 and height 1, with area 3.



Adding the two areas:

$$\int_{-3}^{0} \left(1 + \sqrt{9 - x^2}\right) dx = 3 + \frac{9\pi}{4} \approx 10.07.$$

Question 26

A car's velocity at the beginning of a trip was tracked for one hour, with the following results.

t (hrs)	0	0.2	0.4	0.6	0.8	1
$v(t)$ (km/hr)	10	30	30	70	90	90

(a) Use a right sum to estimate the distance travelled by the car between $t = 0$ and $t = 1$ hour. Give units in your answer.

(b) A car's mileage or gas consumption depends on its speed. For the same car as in part (a), the table below shows the relationship between the speed v and the resulting gas consumption (reported in liters required to travel 100 km).

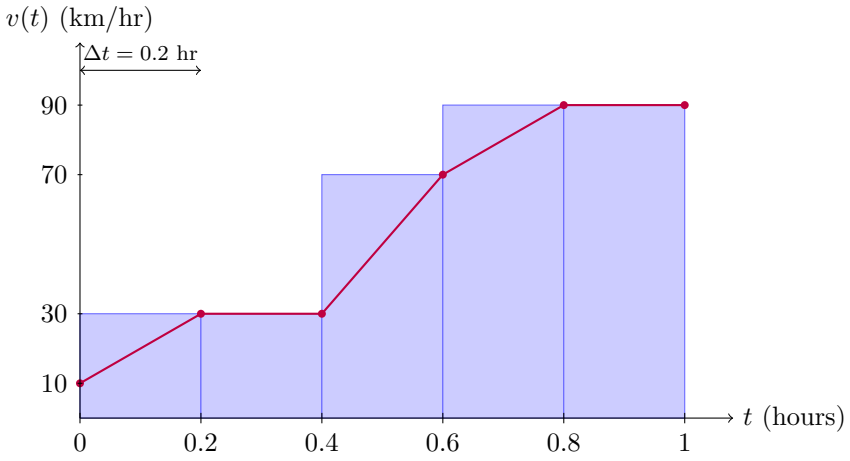
v (km/hr)	10	30	50	70	90	110
Efficiency (liters per 100 km)	29	15	10	7	8	9

Using a right-sum approach for each time interval, use this efficiency information and the trip information from part (a) to estimate the amount of gas consumed by the car between $t = 0$ and $t = 1$ hour. Give units in your answer.

Solution

Part (a). A right sum evaluates the function at the right endpoint of each subinterval:

$$\text{Right Sum} = \sum_{i=1}^{n} f(x_i)\,\Delta x, \qquad \Delta x = \frac{b-a}{n}.$$



Here $n = 5$ and $\Delta t = \dfrac{1-0}{5} = 0.2$ hr. By dimensional analysis,

$$\text{time (hr)} \times \text{velocity } \left(\tfrac{\text{km}}{\text{hr}}\right) = \text{distance (km)}.$$

Using the velocities at $t = 0.2, 0.4, 0.6, 0.8, 1$:

$$\begin{aligned}\text{Distance} &\approx (0.2\text{ hr})(30) + (0.2\text{ hr})(30) + (0.2\text{ hr})(70) + (0.2\text{ hr})(90) + (0.2\text{ hr})(90)\\ &= (0.2\text{ hr})\,(30+30+70+90+90)\ \tfrac{\text{km}}{\text{hr}}\\ &= (0.2\text{ hr})\,\left(310\ \tfrac{\text{km}}{\text{hr}}\right)\\ &= 62\text{ km}\end{aligned}$$

Part (b). By dimensional analysis, we want liters of gas used:

$$\text{Gas (L)} = \text{Efficiency}\left(\tfrac{\text{L}}{100\text{ km}}\right) \times \text{Velocity}\left(\tfrac{\text{km}}{\text{hr}}\right) \times \text{Time (hr)}.$$

For each subinterval, we use the velocity at the right endpoint (Table 1) and its corresponding efficiency (Table 2):

Time (hr)	0.2	0.4	0.6	0.8	1
Velocity (km/hr)	30	30	70	90	90
Efficiency (L/100 km)	15	15	7	8	8
Gas used (L)	$(0.2)(30)\frac{15}{100}$ $= 0.9$	$(0.2)(30)\frac{15}{100}$ $= 0.9$	$(0.2)(70)\frac{7}{100}$ $= 0.98$	$(0.2)(90)\frac{8}{100}$ $= 1.44$	$(0.2)(90)\frac{8}{100}$ $= 1.44$

Therefore,

$$\text{Total gas used} \approx 0.9 + 0.9 + 0.98 + 1.44 + 1.44 = 5.66 \text{ L}.$$

Alternative Solution (Part b)

Another method is to use the distance travelled over each interval from part (a), and multiply by the efficiency corresponding to that segment's velocity. By dimensional analysis,

$$\text{distance (km)} \times \text{efficiency}\left(\tfrac{\text{L}}{100\text{ km}}\right) = \text{gas (L)}.$$

The distance travelled on each interval is $(0.2 \text{ hr})\, v$ (the right-sum terms from part (a), which add up to 62 km):

Time (hr)	0.2	0.4	0.6	0.8	1
Distance (km)	$(0.2)(30) = 6$	$(0.2)(30) = 6$	$(0.2)(70) = 14$	$(0.2)(90) = 18$	$(0.2)(90) = 18$
Efficiency (L/100 km)	15	15	7	8	8
Gas used (L)	$6 \cdot \frac{15}{100}$ $= 0.9$	$6 \cdot \frac{15}{100}$ $= 0.9$	$14 \cdot \frac{7}{100}$ $= 0.98$	$18 \cdot \frac{8}{100}$ $= 1.44$	$18 \cdot \frac{8}{100}$ $= 1.44$

Therefore,

$$\text{Total gas used} \approx 0.9 + 0.9 + 0.98 + 1.44 + 1.44 = 5.66 \text{ L},$$

which agrees with the first method.

Question 27

Consider the integral representing the area under the graph of $f(x) = \cos(x)$ from $x = 0$ to $x = \pi/2$.

(a) Use the trapezoidal rule with four intervals to estimate this area.

(b) Sketch the graph and your trapezoidal estimate of the area. Is your estimate an underestimate or an overestimate of the exact integral value?

Solution

Part (a). The trapezoidal rule approximates the area of each subinterval by a trapezoid:

$$\text{area} = (\text{average of the two heights})(\Delta x) \quad \Longrightarrow \quad \text{Trap}(n) = \sum_{i=1}^{n} \left(\frac{f(x_{i-1}) + f(x_i)}{2} \right) \Delta x.$$

Step 1: Interval width. With $n = 4$,

$$\Delta x = \frac{b-a}{n} = \frac{\pi/2 - 0}{4} = \frac{\pi}{8}.$$

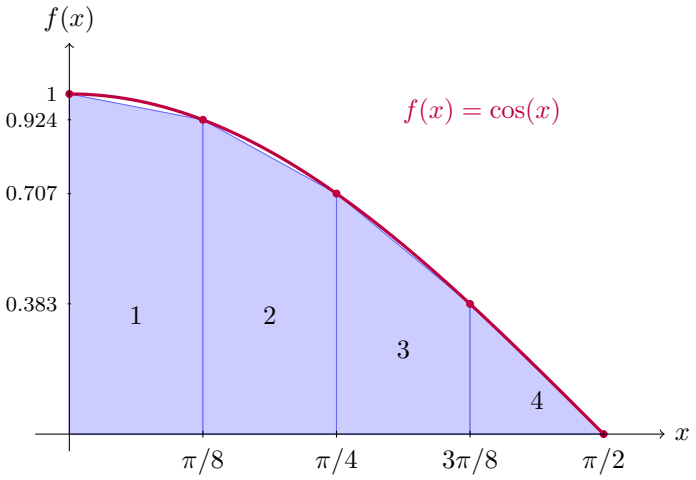
The four intervals are $\left[0, \frac{\pi}{8}\right]$, $\left[\frac{\pi}{8}, \frac{\pi}{4}\right]$, $\left[\frac{\pi}{4}, \frac{3\pi}{8}\right]$, $\left[\frac{3\pi}{8}, \frac{\pi}{2}\right]$.

Step 2: Function values at each endpoint.

$$\begin{aligned} \cos(0) &= 1 \\ \cos\left(\tfrac{\pi}{8}\right) &\approx 0.924 \\ \cos\left(\tfrac{\pi}{4}\right) &= \tfrac{\sqrt{2}}{2} \approx 0.707 \\ \cos\left(\tfrac{3\pi}{8}\right) &\approx 0.383 \\ \cos\left(\tfrac{\pi}{2}\right) &= 0 \end{aligned}$$

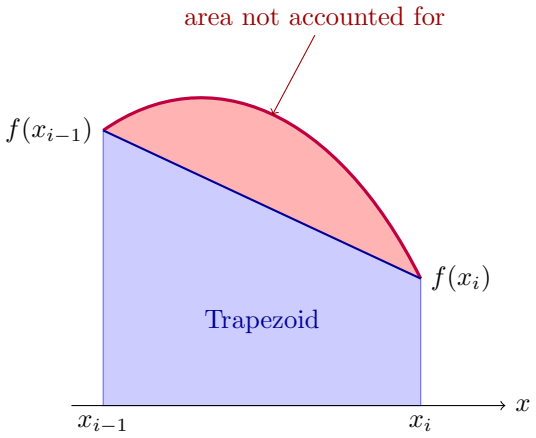
Step 3: Apply the trapezoid estimate.

$$\begin{aligned} \text{Trap}(4) &= \left(\frac{1 + 0.924}{2}\right)\left(\frac{\pi}{8}\right) + \left(\frac{0.924 + 0.707}{2}\right)\left(\frac{\pi}{8}\right) + \left(\frac{0.707 + 0.383}{2}\right)\left(\frac{\pi}{8}\right) + \left(\frac{0.383 + 0}{2}\right)\left(\frac{\pi}{8}\right) \\ &= \left(\frac{\pi}{8}\right)(0.962 + 0.816 + 0.545 + 0.192) \\ &\approx 0.987 \quad \text{(computed using unrounded values of cos)} \end{aligned}$$



Part (b). The trapezoidal estimate is an **underestimate**.

Cosine is concave down on $\left[0, \frac{\pi}{2}\right]$ (since $f''(x) = -\cos(x) \leq 0$ there). Thus the straight line from $f(x_{i-1})$ to $f(x_i)$ on each subinterval lies fully below the curve, so there is a small area above each trapezoid that is not being accounted for.



(Schematic: the curvature is exaggerated for clarity.)